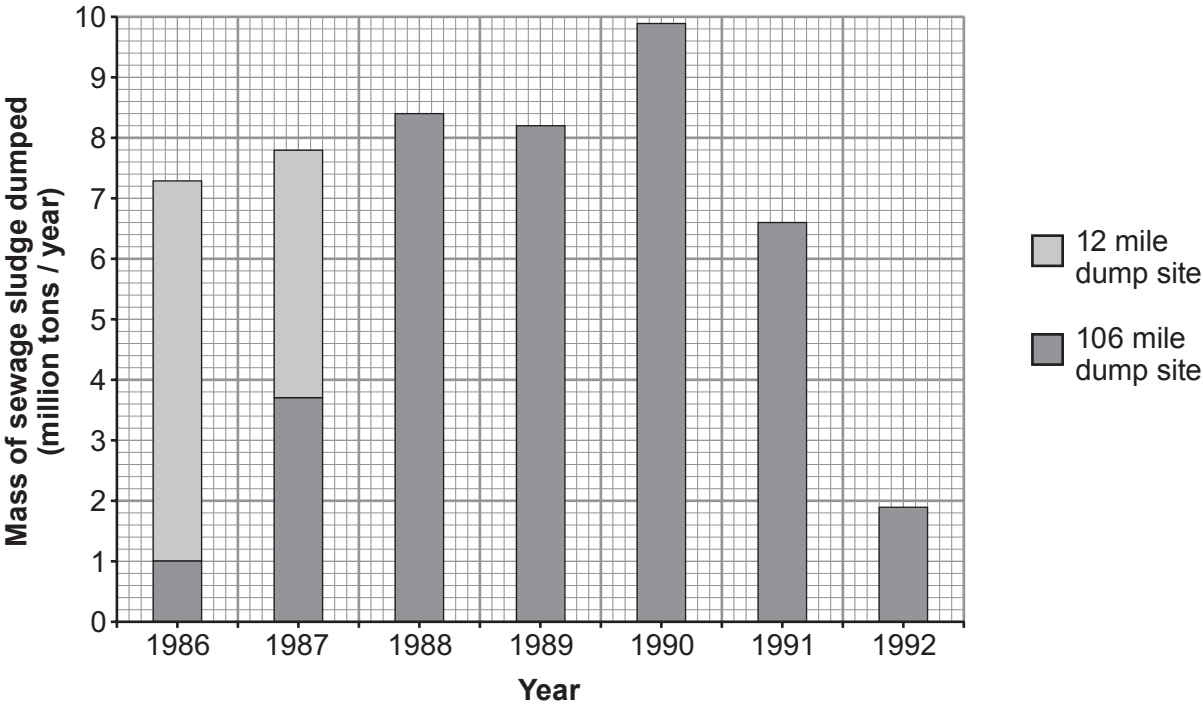


Examiner
only

Graph showing the annual disposal of sewage sludge at the '12 mile' and '106 mile' dump sites from 1986 to 1992.



- (a) (i) Calculate the total mass of sewage sludge disposed of at the '12 mile' dump site in 1986 and 1987. [2]

total mass of sewage sludge = million tons

- (ii) What was the final year in which sewage sludge was disposed of at the '12 mile' dump site? [1]

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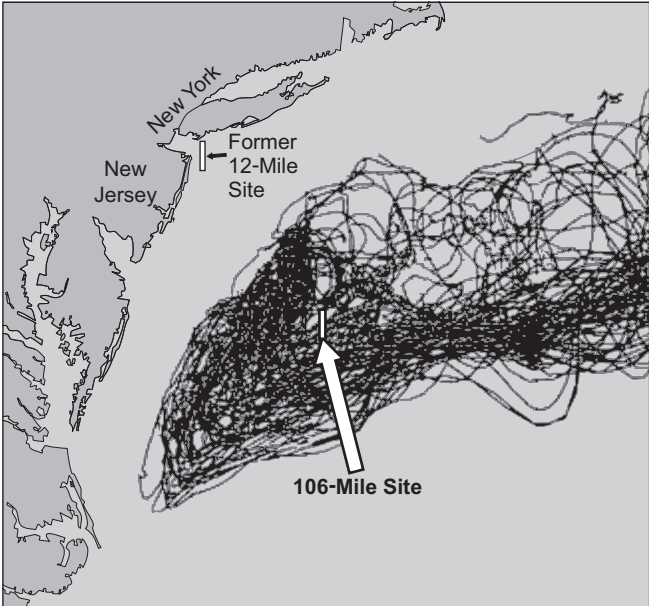


- (iii) The US Environmental Protection Agency released buoys into the ocean at the 106 mile dump site. They used satellites to track the movement of the buoys between 1989 and 1992.

drifting buoy tracked
by satellite



Map 2 showing the movement of buoys



Use the information in map 2 to suggest why it was decided to select a sewage dump site 106 miles off the east coast of the USA and to close the ‘12 mile’ dump site. [2]

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Examiner
only

(b) Of the sewage sludge which is dumped, 20 – 70% reaches the sea bed. Here the oxygen consumed by living organisms increases greatly. Explain why this happens. [3]

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(c) Close to the ‘106 mile’ sewage dump site is the former ‘106 mile’ **industrial** waste dump site. (See map 1 on page 16). Name **one** group of industrial wastes which you would expect to find at this site. [1]

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9

END OF PAPER

